



Effect of rotating magnetic field on the diffusivity of ethylammonium nitrate ionic liquid confined between micrometer-spaced glass plates

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ABSTRACT

We studied changes in the diffusion coefficients in layers of EAN confined between glass plates, placed in a strong magnetic field with a magnetic flux density B_0 , and rotated around the axis directed along and normal to B_0 . Under the rotational conditions along B_0 , the diffusion coefficient decreases with time after placement in the magnetic field at the same rate as for a static sample, observed previously (Filippov and Antzutkin, Phys. Chem. Chem. Phys. 2018, 20, 6316). However, when the EAN layers are rotating around the axis perpendicular to B_0 , the duration of exposure to the magnetic field does not affect the diffusion coefficient until the rotation stops. On the other hand, the diffusivity after extended exposure to a static magnetic field increases as the sample starts to rotate around the axis perpendicular to B_0 . The observed effects are due to either the periodic change in the orientation of the thin surface layers of EAN or inhomogeneity of B_0 in the sample due to the B_0 fluctuation.

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1. Introduction

A magnetic field causes changes in the physicochemical properties of non-magnetic liquid substances such as water [1,2], aqueous solutions [1], nanofluids [3], liquid crystals [4], and ionic liquids [5–7], among others. When water is placed between strong magnets, its conductivity [8], viscosity [9], evaporation [10] and melting temperature [11] increase, and its surface tension varies with the magnetic field strength [12]. The properties of water are very closely connected to the continuous network of hydrogen bonds formed by the water molecules. Therefore, various effects of a magnetic field are related to changes in the hydrogen-bonding network [1,9,13–17]. Changes in intermolecular interactions and electronic perturbations caused by variations in hydrogen bonds can be observed from ^1H NMR chemical shifts [18–20].

Ionic liquids (ILs) are a new class of materials prepared from organic cations and either organic or inorganic anions that remain in a liquid state near room temperature [21–23]. Ethylammonium nitrate (EAN), the most studied IL [22,24], is used as a reaction medium, a precipitating agent for protein separation [25], and an electrically conductive solvent in electrochemistry [26]. EAN has three readily exchangeable protons

on the $-\text{NH}_3$ group of cations; therefore, it belongs to a class of so-called “protic” ionic liquids, which are formed by proton transfer from an equimolar combination of a Brønsted acid and a Brønsted base [22]. Like water, EAN has a three-dimensional hydrogen-bonding (H-bonding) network, which is highly connected, dense, complex and variable [27,28]. H-bonding in ILs is a secondary (after ionic bonding) structure-directing contributor [28]. The presence of the H-bonding network, as with water, affects the physicochemical properties of EAN. The microstructure of protic ILs depends greatly on the hydrogen bond interactions [18].

ILs placed in micrometer-scale confinement [29–35], particularly EAN [6,30,33,36,37], have attracted special interest. Magnetic field strength is an important external variable that influences the physical phase state and dynamic properties of the confined EAN [6,7,37,38]. Enhanced diffusivity of cations as well as accelerated NMR relaxation of $-\text{NH}_3$ protons have been observed for EAN confined in layers with a thickness of $\sim$ four μm between glass plates [7,33]. The main factors responsible for this effect are suggested to be the H-bond network and the polarity of the charged silicon oxide surface [6,33,37]. Exposure to the strong static magnetic field of an NMR spectrometer leads to a gradual decrease in diffusivity of ions until it reaches an equilibrium value less than that in the bulk EAN [6]. These processes are related to the phase transformations of the ILs [6,7] through changes in the H-bonding network, which lead to certain chemical shift changes of the NMR spectra lines [20]. We performed temperature measurements of chemical shifts

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and diffusion coefficients for EAN in the bulk phase (I_B) and confined in layers between glass plates unexposed (I) and exposed (I_{MF}) to a static magnetic field of 9.4 T [20]. These experiments showed that the NMR chemical shift of $-NH_3$ protons of EAN in the I phase is strongly shifted upfield, ~ 0.0145 ppm/K, which was explained by weakening of the hydrogen-bonding network of EAN [20]. Exposure to the static magnetic field leads to restitution of the H-bonding network in the I_{MF} phase. Application of a static magnetic field in directions along and normal to the EAN layers leads to the same magnitude and temporal parameters of the effects of a static magnetic field on diffusivity and NMR relaxation. Decreasing in the external magnetic field to zero (removing the sample from the magnetic field) leads to recovery of the dynamic characteristics of EAN to values prior to the start of the experiment.

The influence of magnetic field parameters on the dynamics and phase state of ionic liquids remains of interest. In this work, we further investigate effects of a magnetic field by rotating an anisotropic sample of EAN confined between glass plates in different directions relative to a strong static magnetic field.

2. Materials and methods

2.1. Sample preparation

Fig. 1 shows the chemical structures of ions of ethylammonium nitrate. EAN was synthesized and characterized at Chemistry of Interfaces of Luleå University of Technology [33]. The content of water in the synthesized EAN was less than 0.05 wt% as determined by Karl-Fisher titration (Metrohm 917 Karl Fisher Coulometer with HYDRANAL reagent).

Samples were prepared with thin borosilicate glass plates ($0.1 \times 2.5 \times 14$ mm), Thermo Scientific Menzel-Gläser, Menzel GmbH, Germany. The plates were arranged in a stack (Fig. 2).

The glass plates were carefully cleaned before use. A stack of glass plates filled with EAN was prepared in a glove box in a dry nitrogen atmosphere. Samples were prepared by adding one μ L of EAN to the first glass plate, placing a second glass plate on top, adding one μ L to this glass plate, etc. until the thickness of the stack reached approximately 2.5 mm. Excess EAN from the sides of the stack was removed by wiping. The sample consisted of a stack of ca 37 glass plates with EAN between them, placed in a rectangular, sealed glass tube. The mean spacing between glass plates was estimated by weighing the introduced EAN, which yielded $d \sim 3.8\text{--}4.5$ μ m [33]. A detailed description of the sample preparation and characterization has been reported in our previous paper [33]. To allow equilibration of ILs inside the samples, experiments were performed within a time period from a week but not later than one and a half months after the sample preparation.

2.2. Pulsed-field gradient NMR diffusometry

1H NMR self-diffusion measurements were performed using a Bruker Ascend/Aeon WB 400 (Bruker BioSpin AG, Fällanden, Switzerland) NMR spectrometer with a resonance frequency of 400.27 MHz for 1H (magnetic field strength of 9.4 T with magnetic

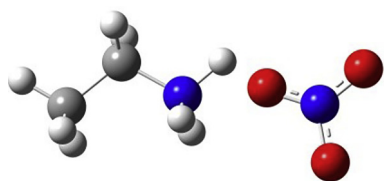


Fig. 1. Chemical structure of EAN: the ethylammonium (EA) cation (left) and the nitrate anion (right) with nitrogen (blue), oxygen (red), carbon (dark grey) and hydrogen (light grey) atoms shown. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

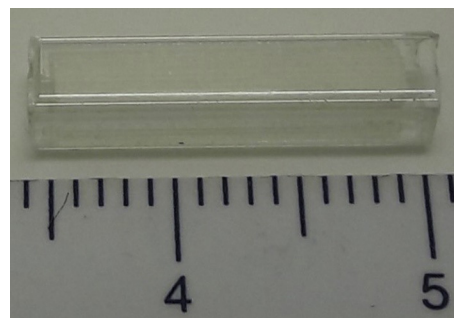


Fig. 2. A sample prepared from 37 glass plates ($0.1 \times 2.5 \times 14$ mm) with confined EAN for NMR diffusion measurements. The ruler below is scaled in cm units.

field homogeneity better than $2.3 \cdot 10^{-7}$ T) using a Diff50 Pulsed-Field-Gradient (PFG) probe. The homogeneity of the magnetic field was estimated from the 1H NMR line width. The diffusional decays (DD) were recorded using the stimulated echo (StE) pulse trains. For single-component diffusion, the form of the DD can be described as [39,40]:

$$A(\tau, \tau_1, g, \delta) \propto \exp\left(-\frac{2\tau}{T_2} - \frac{\tau_1}{T_1}\right) \exp\left(-\gamma^2 \delta^2 g^2 D t_d\right) \quad (1)$$

Here, A is the integral intensity of the NMR signal, τ and τ_1 are the time intervals in the pulse train; γ is the gyromagnetic ratio for protons; g and δ are the amplitude and the duration of the gradient pulse; $t_d = (\Delta - \delta/3)$ is the diffusion time; $\Delta = (\tau + \tau_1)$. D is the diffusion coefficient. In the measurements, the duration of the 90° pulse was 7 μ s, δ was in the range of $(0.5 \div 2)$ ms, τ was in the range of $(3 \div 5)$ ms, and g was varied from 0.06 up to 29.73 T·m $^{-1}$. Diffusion time t_d was varied from 4 to 500 ms. The recycle delay during accumulation of signal transients was 3.5 s. Data were processed using Bruker Topspin 3.5 software. Non-linear least squares regression was used to fit the experimental data with Eqs. (1) to extract D values. The temperature of the sample was 294 K.

The sample under study is anisotropic. For the flat geometry of EAN layers, two directions are principally important: normal to the plates and any direction perpendicular to n , the direction along the EAN layers. Therefore, we studied the effects of rotation of the sample axis directed along plates relative to two axis: along and normal to the static magnetic field B_0 of the NMR spectrometer. A schematic of the experiment is shown in Fig. 3.

In the first arrangement (Fig. 3A), the sample was connected to the standard 5-mm NMR tube spinner. In this case, the sample was rotated around the z axis, parallel to the main static magnetic field B_0 . The angle between n and the axis of rotation did not change during rotation and

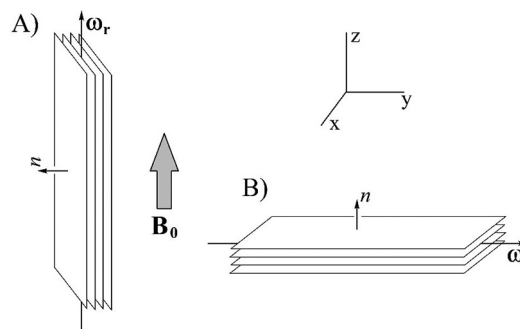


Fig. 3. A schematic of the experiment with rotation of the sample (glass plates filled with EAN) in two different directions relative to the static magnetic field of the NMR spectrometer, B_0 . ω_r is the axis of rotation of the sample. n is the direction normal to the glass plates. x , y and z are axes in the laboratory coordinate system.

was equal to 90 degrees. EAN layers were always directed along the axis of rotation. The rate of rotation was ~ 20 Hz. In the second arrangement, an NMR solenoid ^1H insert was used to macroscopically align the plates (Fig. 3B). We used an assembly made of solenoid ^1H insert (the same that previously used in our previous work [6,7]), a type of flexible transmission, gear and a synchronous AC geared reversible one-phase motor (240 V, 30 rpm, 9.5 W). In this case, the sample was rotated around the y-axis directed perpendicular to the direction of the main static magnetic field $\mathbf{B}_0$. The angle between n and $\mathbf{B}_0$ is changing during rotation, therefore the EAN layers are changing their orientation during the experiment. The rate of rotation was $1/3 \text{ min}^{-1}$ ($\sim 5.6 \cdot 10^{-3} \text{ Hz}$). To perform the NMR diffusion measurement, the rotation was stopped for both arrangements (A) and (B) and, in the case of (B); the sample was turned with n perpendicular to $\mathbf{B}_0$. So, mechanical collisions of EAN with the glass plates did not influence the values obtained for the diffusion coefficients, as was shown previously [33].

3. Results and discussion

We performed several experiments with two sample arrangements designated (A) and (B) in Fig. 3. The changes in diffusion coefficients with time in these experiments are shown in Fig. 4.

In the case of the static sample, after placing the sample in the magnetic field, D decreases with time (curve 1, open circles) from D_i to D_{MF} for both, (A) and (B) arrangements of the sample, as was shown earlier [6,37]. For arrangement (A) when EAN layers are rotating around the axis parallel to $\mathbf{B}_0$, diffusion coefficient decreases with time just after placement in the magnetic field (curve 2, solid circles) at the same rate as for a static sample from D_i to D_{MF} . After the diffusivity has reached D_{MF} , the diffusivity does not change anymore until the sample is removed from the field. Thus, the effect of rotating the sample in the magnetic field around the axis parallel to $\mathbf{B}_0$ is similar to that obtained previously for the static sample [6].

In the case of arrangement (B), when the EAN layers are rotating around the axis normal to $\mathbf{B}_0$, rotation of the sample for 18 h after sample placement in the magnetic field does not affect the diffusion coefficient (curve 3, solid triangles), and remains nearly equal to D_i during the time of the experiment. However, after the rotation is stopped, D starts

to decrease in the same manner as occurs with the static sample and with the rotating sample in the case of (A) (curve 4, open triangles), and finally reaches D_{MF} . However, for the static sample with the same orientation (B) effect of the static magnetic field is the same as for the static sample of arrangement (A) [6]. Therefore, rotation of the EAN layers around the axis normal to $\mathbf{B}_0$ stops the effect of a static magnetic field. Diffusivity of confined EAN does not change under the influence of a magnetic field under the latter conditions.

One more experiment was performed with the sample under arrangement (B), which previously reached equilibrium in the static magnetic field and its diffusion coefficient stabilized near D_{MF} . Rotation of the sample around the axis normal to $\mathbf{B}_0$ started simultaneously with measurements. During this time, the diffusion coefficient increased (curve 5, half-filled triangles). As is shown, the value of D tended toward D_i and reached a value in the middle of the values for D_{MF} and D_i . The recovery curve obtained for static sample after its removing from the magnetic field [37] is shown by dashed line. Therefore, rotation of the sample reduces the effect of a static magnetic field on diffusivity of EAN.

Each experiment was repeated at least twice. After the end of each experiment, the sample was removed from the magnet for a week to completely restore phase I.

In further discussion, we will use the same notations for dynamic EAN phases as we used earlier [7]: “ $\mathbf{I}_b$ ” is the bulk phase, “ $\mathbf{I}$ ” is the phase formed in confinement in the absence of the magnetic field and “ $\mathbf{I}_{MF}$ ” is the phase formed after exposure to the magnetic field without rotation of the sample.

A pertinent question is: Which mechanisms control changes of diffusivity of confined EAN due to rotation? Indeed, under rotation of the sample, additional forces may influence the phase structure and corresponding dynamics of confined EAN. Firstly, this is the inertia force, which is maximal for rotation with a frequency of ~ 20 Hz that occurs in the case of the sample rotating around $\mathbf{B}_0$ (arrangement (A) in Fig. 3). From the obtained results (Fig. 4, curve 2), it is evident that the contribution of this force is negligible, because the decrease in D (curve 2) coincides with that obtained for the sample without rotation. Obviously, the effect of the force is also negligible in cases when the sample is rotating with lower a frequency, $\sim 5.6 \cdot 10^{-3} \text{ Hz}$ (arrangement (B), curves 3,5). Therefore, at the experimental conditions, rotation of the sample in the static magnetic field is equivalent to the rotation of magnetic field for the static sample.

Secondly, EAN, as any other ionic liquid, is partly dissociated into ions. The ratio of the molar conductivity obtained from the impedance measurements (Λ_{imp}) and the molar conductivity calculated from the diffusion coefficients of ions (Λ_{NMR}) is commonly used to investigate the ionic association and dissociation under equilibrium in RTILs. The $\Lambda_{imp}/\Lambda_{NMR}$ indicates the percentages of ions responsible for the ion conduction within the diffusing unit and, therefore, called as “ionicity” in the case of ionic liquids [41]. The molar conductivity of the neat IL is calculated from the self-diffusion coefficients using the Nernst-Einstein Eq. [42]:

$$\Lambda_{NMR} = F^2(D_{cation} + D_{anion})/RT \quad (2)$$

where F is the Faraday constant and R is the universal gas constant. Ionicity of bulk EAN at 298 K was estimated taken diffusion coefficients of cation ($4.3 \cdot 10^{-11} \text{ m}^2/\text{s}$) and anion ($6.5 \cdot 10^{-11} \text{ m}^2/\text{s}$) [33], and $\Lambda_{imp} = 2.1 \text{ mS}\cdot\text{cm}^2\cdot\text{mol}^{-1}$ [43]. The estimation gives 0.35 for the ionicity. Therefore, there are high fractions of free charges in bulk EAN, which can be subjected to external magnetic field. Unfortunately, we cannot calculate ionicity of confined EAN because of lacking data of its electrical conductivity data. However, we can suggest that it does not differ much, as a rough approach.

Due to displacements of charges in a magnetic field, Lorentz forces might appear which can also affect dynamic processes occurring in the IL. Evidently, the effect of Lorentz forces is also negligible in our

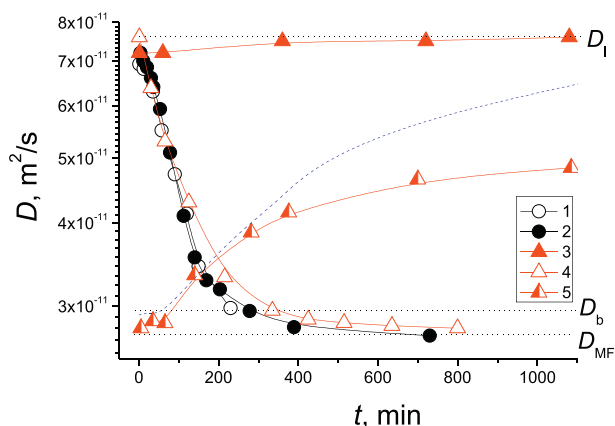


Fig. 4. Evolution of diffusion coefficient of the ethylammonium cation of EAN confined between glass plates in a magnetic field of 9.4 T/m for arrangements (A) and (B) with axes of rotation oriented along (circles) and normal (triangles) to $\mathbf{B}_0$, respectively. 1, 4 are static experiments. 2 corresponds to rotation with frequency $\omega_r \sim 20$ Hz. 3, 5 correspond to $\omega_r \sim 5.6 \cdot 10^{-3} \text{ Hz}$. The time scale started at placing samples in the magnetic field (1,2,3), after stopping rotation (4) and at the start of rotation of the sample extensively exposed to the magnetic field (5). Data 1 was reproduced from [6,36]. Diffusion coefficients D_b , D_i and D_{MF} correspond to equilibrium values for bulk EAN, confined between glass plates unexposed and extensively exposed to a magnetic field of 9.4 T/m, respectively. Dashed line shows recovery of diffusivity from D_{MF} to D_i for the static sample [37]. $T = 294 \text{ K}$.

study, because experiments with a static sample and a sample rotating around $\mathbf{B}_0$ show similar results.

Characteristic time constants for transformations of dynamic phases of EAN placed between glass plates are on the order of a few hours (curves 1,2,5 in Fig. 4). For example, the diffusivity decreases by a factor of two over approximately 70 min [6], the same as for recovery [37]. For both arrangements used, (A) and (B), rotation of the sample occurs at different rates, ~20 Hz and ~ $5.6 \cdot 10^{-3}$ Hz, but revolution times in both cases are much less than the time-scale of the EAN $\mathbf{I} \rightarrow \mathbf{I}_{\text{MF}}$ and $\mathbf{I}_{\text{MF}} \rightarrow \mathbf{I}$ phase transformations. Formations occurring in the $\mathbf{I}$ and $\mathbf{I}_{\text{MF}}$ phases of the confined EAN originate from EAN interacting with the polar surface of glass plates. Therefore, one of the possible mechanisms leading to the observed effects of sample rotation (Fig. 4) is the change in orientation of the surface layers of EAN interacting with Si-O-H groups near the glass plates, which gives insignificant contribution to the NMR signal, but is nevertheless present [30,31]. For the sample rotating around the axis directed along the magnetic field (arrangement (A)), the orientation of the surface layer does not change and the sample rotation does not influence the kinetics of the decrease in D , which remains typical for a stationary sample (Fig. 4, curves 1 and 2). However, for the sample under arrangement (B), rotation leads to a periodic change in the orientation of the surface layer. Because the interaction of EAN with the surface of the glass plates is responsible for the formation of the $\mathbf{I}$ and $\mathbf{I}_{\text{MF}}$ phases of EAN, the changing orientation of the surface layer of EAN relative to $\mathbf{B}_0$ appears to be a leading factor resulting in preservation of the $\mathbf{I}$ phase (curve 3) or the $\mathbf{I}_{\text{MF}} \rightarrow \mathbf{I}$ transformation (curve 4). The surface layer of EAN may include up to seven molecular layers with the spatial range up to three nm [30]. Subsequent interaction of the surface layer of EAN with other, more distant regions of the IL (~ four μm), not directly interacting with the glass surface, occurs on a slower time scale and leads to modification of the H-bonding network of all of the EAN confined between glass plates [20]. It is known that EAN can form clusters with different structure of the H-bonding network [44]. It is possible that the external influence can change the structure. We do not know how this proceeds in our particular case. Extra simulations are necessary to elucidate how the interactions of EAN with the glass surface and the magnetic field altering the hydrogen-bonding network.

Rotation of the sample in the (B) arrangement may also lead to a variation of magnetic field strength in the EAN layers, because of the difference in magnetic susceptibility of glass and EAN [45]. We estimated this variation of $\mathbf{B}_0$, $\Delta\mathbf{B}$, by taking into account the variation of the ^1H NMR chemical shift $\Delta\delta$ at rotation of the sample from 0 to 90 degrees, using Eqs. [45]:

$$\Delta\omega = \Delta\delta \cdot \omega_0 \quad (3)$$

and

$$\Delta\mathbf{B} = \Delta\omega/\gamma \quad (4)$$

For $\Delta\delta \sim 4$ ppm, the estimation showed that $\Delta\mathbf{B} \sim 4.0 \cdot 10^{-5}$ T. This is comparable with the width of the EAN ^1H NMR spectrum ($\sim 5.4 \cdot 10^{-5}$ T). Therefore, alteration of the magnetic field during the sample rotation may also prevent $\mathbf{I} \rightarrow \mathbf{I}_{\text{MF}}$ transformation or it can cause $\mathbf{I}_{\text{MF}} \rightarrow \mathbf{I}$ transformation for the sample in arrangement (B). In this case, the H-bonding network should be involved in the transformation.

4. Conclusions

We studied variations of diffusion coefficients in layers of EAN confined between glass plates exposed to a strong magnetic field, with rotation of these layers around the axis directed along the lines of magnetic flux $\mathbf{B}_0$ and normal to it. In the first case, the diffusion coefficient decreases with time after placement in the magnetic field at the same rate as in the case of a static sample. However, in the case of EAN layers rotating around the axis perpendicular to $\mathbf{B}_0$, rotation

eliminates the effect of the static magnetic field until the rotation is stopped. On the other hand, the diffusivity in the layers reached after extended exposure to the static magnetic field, increases as the sample starts to rotate around the axis perpendicular to $\mathbf{B}_0$. The observed effects of the sample rotation around the axis normal to magnetic field is conditioned by the periodic change in orientation of thin surface layers of EAN interacting with Si-O-H groups near the glass plates relative to $\mathbf{B}_0$, or due to periodic alteration of magnetic field strength in the EAN layers occurring due to a difference in magnetic susceptibility of glass and EAN during rotation of the sample. Subsequent interaction of the surface layers with other parts of EAN, not directly interacting with the glass surface or alternating magnetic field, weaken the hydrogen-bonding network within the confined EAN. The weakening of the hydrogen-bonding network in EAN layers confined between glass plates has been demonstrated by ^1H NMR chemical shift and diffusivity data in our previous work [20].

It is observed from these experiments that the rate of rotation of the sample around the axis normal to the static magnetic field is critical for the H-bonding transformation in confined EAN. Therefore, we planning for further detailed studies to elucidate how the rotation rate influences on the magnitude and on the rate of transformations occurring with confined EAN under the influence of the magnetic field.

CRediT authorship contribution statement

Andrei Filippov: Methodology, Conceptualization, Investigation, Writing - original draft. **Rustam Gimatdinov:** Software, Validation. **Oleg N. Antzutkin:** Writing - review & editing. **Natalia A. Kuzina:** Investigation. **Oleg I. Gnezdilov:** Investigation, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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